

HARSHAL KOLHE

 [GitHub](#) |  [LinkedIn](#) |  harshalkolhe03@gmail.com

EDUCATION

Visvesvaraya National Institute of Technology, India

Bachelor of Technology, Minor in Internet of Things

Aug' 23 – May' 27

CGPA: [8.66/10.0](#)

EXPERIENCE

Indian Institute of Technology, Roorkee

May' 25 – Jul' 25

Summer Research Intern | [\[Certificate\]](#)

- Developed a reinforcement learning-based masystem for a quadcopter carrying a suspended payload, with active suppression of payload-induced oscillations during flight.
- Built a high-frequency data collection and processing pipeline that synchronized 20+ multi-source flight and 8+ payload unit data streams for reinforcement learning policy validation.
- Configured a real-time autonomy stack using a Raspberry Pi as the onboard flight computer, streaming flight-state variables from CubeOrange at 100 Hz via MAVLink, performing onboard RL inference, and generating flight-control commands for control.

PROJECTS

Context Aware Navigation Assistant | [\[GitHub\]](#)

May' 26 – Jul' 26

- Designed an end-to-end autonomous perception and navigation system, capable of understanding dynamic traffic environments, evaluating risks, and generating real-time navigation decisions for safe autonomous operation.
- Built a computer vision pipeline using YOLOv8 and OpenCV to detect vehicles, pedestrians, traffic signals, and road signs in traffic scenes, achieving 75% accuracy and 90% precision.
- Implemented spatial reasoning and risk assessment modules to evaluate object proximity, safe turning opportunities, and navigation actions such as stop, proceed, and turn, achieving 8% correct action decisions.

Multiagent Hybrid Disaster Rescue Drones | [\[GitHub\]](#)

Jan' 25 – Apr' 26

- Developed a two-drone system that searches in a lawnmower pattern, detects humans, and diverts to deliver relief packages.
- Fine-tuned and deployed a YOLO object detection model for aerial imagery, integrating a real-time RTSP video pipeline from the SIYI A8 Mini using OpenCV to process live frames and detect targets, achieving 85% detection accuracy at 7 m altitude.
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Dual UAV Scan and Spray Architecture for Precision Agriculture | [\[GitHub\]](#)

Jul' 25 – Dec ' 25

- Engineered a real-time crop-disease detection pipeline using OpenCV and HSV-based image segmentation, employing dual-mask filtering and ND-filter preprocessing to achieve 70% detection accuracy at 6 m altitude under varying lighting conditions.
- Created a real-time plant-cluster detection and classification pipeline to determine nozzle diameter and spraying parameters, integrating vision outputs with a Raspberry Pi-based control system for automated spraying.
- Integrated the vision and control pipeline with a custom carbon-fiber quadcopter capable of carrying a 10 kg fluid payload, achieving 25 cm spraying accuracy during field testing.

PUBLICATIONS

- S. Kshirsagar, **H. Kolhe**, R. Deshmukh, K. Ayush, S. Dhalwar, S. Biswas, M. Limaye, and S. S. Chiddarwar, "A Multi-Agent Drone System for Real-Time Victim Detection and Aid Delivery," presented at the International Conference on Advances in Mechanical and Manufacturing Systems (ICAMMS), VNIT Nagpur, India, 2026.
- A. Sharma, M. Ravikiran, S. Chakrabarty, R. Saluja, M. Limaye, **H. Kolhe**, and S. Samiron, "Prompted to Fly: Translating Free-Form Instructions into Schema-Constrained Mission Generation for UAVs using LLMs," 2025 IEEE International Conference on Big Data (BigData), Macau, China, 2025, pp. 7278-7286, doi: [10.1109/BigData66926.2025.11401432](https://doi.org/10.1109/BigData66926.2025.11401432).

SKILLS

Languages: C, C++, Python

Softwares and Tools: ArduPilot, QGroundControl, Mission Planner, Linux, ROS2

Libraries: OpenCV, Ultralytics (YOLOv8), NumPy, Pandas, Matplotlib, DroneKit, MAVSDK, Mavlink

EXTRACURRICULAR ACTIVITIES

- Secured the [Best Paper Presentation Award](#) at the ICAMMS Conference for presenting research work on autonomous systems.
- Secured the Ignite Prize in **NIDAR**, a national-level UAV autonomy Competition, placing among the top 5 out of 100+ teams.
- Secured [First Runner-up](#) prize at the Drone Helix Competition, SVNIT Surat.
- Vice Chairman of IvLabs, Robotics and Artificial Intelligence Lab at VNIT, Nagpur.